# Memory and Forecasting

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July 8, 2026

## 1 Forecasting in Economics

Across a wide range of economic settings, forecasters systematically overweight recent information. Analysts who observe strong recent earnings revise long-term growth forecasts upward by more than the data warrant (La Porta, 1996; Bordalo et al., 2019). Credit spreads overreact to recent default experience (Greenwood and Hanson, 2013). Investors extrapolate from recent stock returns, growing optimistic after booms and pessimistic after downturns (Greenwood and Shleifer, 2014). In macroeconomic surveys, individual forecasters revise their beliefs too sharply in the direction of their most recent private signals (Bordalo et al., 2020)—even as consensus forecasts exhibit sluggish adjustment to aggregate shocks, a pattern that emerges naturally when individual overreaction is averaged across forecasters with dispersed information (Coibion and Gorodnichenko, 2015; Woodford, 2003). The resulting biases can generate excess volatility in financial markets (Shiller, 1981), boom-bust cycles in credit and investment (Gennaioli et al., 2016), and persistent forecast errors that complicate the conduct of monetary policy. The phenomenon is well documented. The question is *why* recent information receives such disproportionate weight.

Several theoretical frameworks capture this recency bias, but none derive it from deeper cognitive principles. Barberis et al. (1998) show that the representativeness heuristic can lead investors to perceive trending regimes in random sequences, generating overreaction to streaks of consistent news. Bordalo et al. (2018) develop this insight in their model of *diagnostic expectations*, where agents overweight states of the world that have become more likely in light of recent news—a formalization of representativeness (Tversky and Kahneman, 1974). Fuster et al. (2010) propose a model of *natural expectations* in which agents use a misspecified model that overfits to recent data. Extrapolative asset pricing models assume that beliefs are an exponentially weighted average of past returns, with a fixed recency parameter calibrated from survey data (Barberis et al., 2015, 2018). Each of these frameworks describes the same bias—disproportionate weighting of recent observations—but treats it as a primitive rather than deriving it from the cognitive processes that give rise to it. In a comprehensive survey of psychology-based asset pricing models, Barberis (2018) identifies several candidate microfoundations for extrapolation, including representativeness, bounded rationality, and simple reinforcement, but notes that no single mechanism has been empirically established as dominant. The survey also highlights the emerging literature on *experience effects*: investors who personally lived through periods of high stock returns subsequently hold more optimistic beliefs and allocate more to equities than investors who did not (Malmendier and Nagel, 2011). The finding that *how* information was experienced—not merely what information was available—shapes subsequent beliefs points directly toward the encoding and retrieval processes that are central to memory science.

Wachter and Kahana (2024) proposed a *retrieved-context theory* of financial decisions that provides the missing microfoundation. In their model, when investors form expectations about future returns, they retrieve past experiences from memory, and the accessibility of those experiences depends on the similarity between the current mental context and the context in which each experience was originally encoded. Because context drifts with ongoing cognitive activity, recent experiences are encoded in contexts more similar to the present, making them more accessible at retrieval. The resulting recency-weighted belief, in equilibrium, generates the patterns of overreaction and excess volatility documented in financial data. The overweighting of recent information is not an assumed bias—it is an emergent consequence of how human memory works. If this claim is correct, then the tools and theories of memory science—developed over more than a century of experimental research—should be directly informative about the origins, magnitude, and boundary conditions of forecasting biases. Yet despite the centrality of expectations to economic theory, there has been remarkably little direct experimental work connecting the cognitive processes underlying expectation formation to the theoretical and empirical literatures on human memory. The present paper begins to bridge this gap.

Afrouzi et al. (2023) take an important step toward bringing the overreaction phenomenon into the laboratory. Their participants forecast the next value of a simple autoregressive process, and the resulting data reveal clear evidence of overreaction: people place too much weight on the most recent observation, particularly when the process is only weakly persistent. The paradigm was not designed to test memory-based accounts, however, and its key design feature—a continuous graphical display of the entire time series—makes it impossible to determine whether the observed bias reflects memory retrieval, visual attention, or some combination of the two. As we argue in Section 2, a paradigm of a rather different kind is needed to test the claim that overreaction is fundamentally a memory phenomenon.

The retrieved-context framework provides two distinct channels through which memory can distort economic judgments, corresponding to the two fundamental modes of memory retrieval: *recall* and *recognition*. The recall channel, formalized in Wachter and Kahana (2024), generates overreaction through a recency gradient: because temporal context drifts, recently encoded observations are retrieved more strongly, producing forecasts that overweight recent values. The recognition channel, formalized in Kahana et al. (2025), generates distorted probability judgments through a different mechanism: the contextual signatures retrieved by a cue and its target are compared, and "decoy" associations erode the similarity between them, producing judgments consistent with base-rate neglect and the representativeness heuristic (Kahneman and Tversky, 1972). These two channels respond differently to experimental manipulation: recency in recall is governed by temporal contiguity and is sensitive to the amount of intervening activity between encoding events, whereas recognition-based similarity judgments depend on the structure of associations across experiences and are less sensitive to temporal spacing (Osth and Fox, 2019). This dissociation provides the theoretical leverage for our experiment: manipulations of temporal spacing should modulate the recall-based component of overreaction while leaving the recognition-based component largely intact. Our approach complements that of Barberis and Jin (2026), who ground overreaction in the interaction between model-free reinforcement learning and model-based belief updating. The two frameworks locate the source of overreaction in different cognitive systems, and our experimental manipulations of temporal spacing and encoding depth generate predictions that can discriminate between them.

Here we investigate, through a series of experiments, whether the overreaction documented in economic forecasting tasks arises from the memory processes described by retrieved-context theory. We adopt the general approach of Afrouzi et al. but adapt it to align with established methodologies in memory science. We eliminate the graphical display, requiring participants to forecast from memory alone. We introduce manipulations of the temporal spacing between observations and the depth of encoding at each observation—manipulations that retrieved-context models predict will modulate the steepness of the recency gradient and hence the magnitude of overreaction. The resulting experiment speaks simultaneously to two literatures: it provides economists with a mechanistic account of a bias documented across dozens of empirical settings, and it provides memory scientists with a consequential real-world domain in which the principles of retrieved-context theory can be tested and extended.

## 2 Memory and prediction

The sensitivity of beliefs to recent information has a long history in experimental psychology. Beginning in the late 1940s, the *probability learning* literature demonstrated that people predicting binary sequences are powerfully sensitive to local sequential dependencies—tracking runs, alternations, and short-range patterns (Estes, 1950; Humphreys, 1939; Grant et al., 1951; Anderson, 1960). Estes (1957, 1964) developed stimulus sampling theory (SST) to account for these results—a framework in which recency weighting emerged as a property of the learning process rather than an assumed parameter, and that was, in essence, an implicit memory model. A parallel line of work on numerical judgment, structurally closer to economic forecasting, told the same story: Anderson (1964) showed that people form estimates as a recency-weighted average of sequential evidence, and Hogarth and Einhorn (1992) demonstrated that sequential belief updating exhibits strong order effects favoring the most recent information. Yet these frameworks—SST, linear operator models (Bush and Mosteller, 1955), information integration theory—all captured recency as a parameter (a learning rate, a recency weight) rather than deriving it from the encoding, storage, and retrieval processes that govern human memory. The missing connection was between the recency effects observed in prediction tasks and those that are among the most robust phenomena in memory research proper—the serial position curve, the recency effect in free recall, and the temporal contiguity effect (Murdock, 1962; Kahana, 1996; Howard and Kahana, 2002).

Retrieved-context models fill this gap. Beginning with the Temporal Context Model (TCM; Howard and Kahana, 2002) and continuing through the Context Maintenance and Retrieval framework (CMR; Polyn et al., 2009; Lohnas et al., 2015), these models produce recency as an emergent property—much as SST did—but replace SST's abstract trace-updating mechanism with an explicit account of how temporal context evolves during encoding and serves as a retrieval cue. The critical advance for the present purposes is that retrieved-context theory transforms the recency weight of classical information integration theory from a free parameter into a quantity with specific, testable determinants: the temporal spacing between observations, which controls the amount of contextual drift between encoding events, and the depth of encoding at each observation, which controls the strength of individual memory traces. These two factors can be manipulated independently in the laboratory.

The Afrouzi et al. (2023) paradigm, reviewed briefly in the introduction, takes the important step of bringing overreaction into a controlled setting—but several features of the design limit its ability to speak to memory-based accounts. Most fundamentally, the entire history of the process remains visible on a graphical display throughout the task, so participants need never retrieve past observations from episodic memory; the forecasting task is closer to a perceptual judgment of a visible pattern than to the memory-based integration process that retrieved-context theory describes. Beyond the display, three additional features constrain what the paradigm can reveal about memory. First, the presentation rate is uniform, with no variation in intervening cognitive activity between successive observations—yet spacing is a primary determinant of the recency gradient under retrieved-context theory. Second, encoding conditions are uncontrolled: participants passively observe each new realization on the graph, with no requirement that they process the precise numerical value; a participant might track the trajectory as a visual gestalt—"it went up, then down"—without encoding the specific values that a memory-based forecast would require. Third, in several experiments forecasts are collected at multiple horizons on each trial, introducing a *retrieval practice* confound: each forecast elicitation is itself a retrieval event that can modify the accessibility of information for subsequent retrievals, making the longer-horizon forecasts difficult to interpret as independent measures of belief.

If overreaction in forecasting is indeed a memory phenomenon, a direct test requires a paradigm in which (a) participants must retrieve past observations from memory rather than reading them from a display, (b) the spacing between observations is experimentally varied to manipulate context drift, (c) the depth of encoding is controlled to manipulate trace strength, and (d) only a single forecast is collected per trial to avoid retrieval-practice confounds. These are the design principles that guide the experiment we describe next.

# 3 New Experiment

We propose an experiment that eliminates the graphical display, requiring participants to forecast from memory alone. The task adopts an *anticipation learning* structure, familiar from the serial learning literature: on each trial the participant first predicts the next value, then sees the actual outcome together with feedback, and finally completes a distractor task of variable duration before the next trial begins (Figure 1). This design transforms the forecasting task into a sequential learning and retrieval paradigm that maps directly onto the theoretical framework of retrieved-context models. The primary manipulation is the temporal spacing between successive observations: drawing on the Context Maintenance and Retrieval (CMR) framework (Polyn et al., 2009; Lohnas et al., 2015), we vary the duration of a cognitively engaging distractor task between observations, thereby controlling the amount of contextual drift and generating predictions about overreaction that are specific to memory-based accounts.

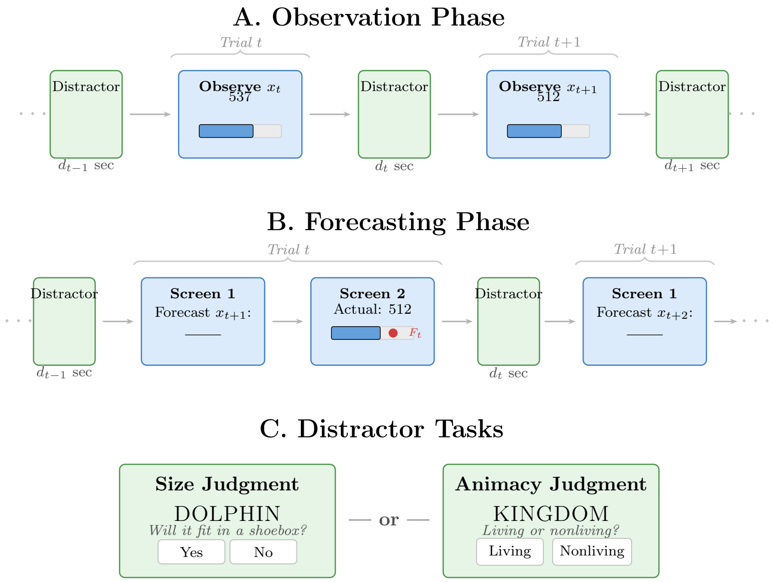


Figure 1: Trial structure for the observation and forecasting phases. **A.** In the observation phase, participants view each value $x_t$ displayed as a number and bar graphic. In the forced-encoding condition, participants type the value before proceeding; in the passive condition, they observe only. A lexical judgment distractor task of variable duration ($d_t \in [2, 12]$ s) separates successive observations. **B.** In the forecasting phase, Screen 1 elicits a one-step-ahead forecast. Screen 2 reveals the actual value on a bar graphic, with a red dot indicating the participant's forecast. Participants in the forced-encoding condition type the actual value. A distractor of variable duration follows, inducing context drift before the next trial. Only one forecast is collected per trial. **C.** Participants are randomly assigned (between subjects) to one of two lexical judgment tasks used as distractors: a size judgment ("Will this item fit in a shoebox?") or an animacy judgment ("Living or nonliving?"). Words are drawn from the 1,638-word pool developed for the Penn Electrophysiology of Encoding and Retrieval Study (Kahana et al., 2024). Values of the AR(1) process are drawn with mean $\mu = 500$ on a 1–1000 scale.

## 3.1 The CMR Account of Overreaction

Under CMR and related retrieved-context models (Howard and Kahana, 2002), each experienced event is encoded in association with a temporal context vector that drifts as a function of ongoing cognitive activity. At retrieval, the current context serves as a cue, and items whose encoding contexts are similar to the current context are retrieved with higher probability and greater strength—the basis of the classic recency effect.

Consider the sequence of events on trial $t$ of the forecasting phase (Figure 1B). The participant enters a forecast for $x_{t+1}$, then sees the actual value on Screen 2 and encodes it (typing the value in the forced-encoding condition). A distractor of duration $d_t$ follows, during which temporal context drifts. When trial $t + 1$ begins, the most recently encoded observation ($x_{t+1}$) is separated from the current moment by the distractor $d_t$; the next-most-recent observation ($x_t$) is separated by $d_t$ plus the within-trial activity of trial $t$ plus the earlier distractor $d_{t-1}$; and so on. The forecast therefore reflects a recency-weighted integration of past observations, where the steepness of the gradient depends on the cumulative context drift between each observation and the current retrieval event—precisely the pattern that Afrouzi et al. (2023) document as "overreaction."

CMR makes a specific quantitative prediction that the Afrouzi et al. paradigm cannot test: the steepness of the recency gradient should depend on the *relative* durations of the distractors separating successive observations. This prediction is closely related to findings in the continuous distractor free recall literature, where the ratio of inter-presentation interval to retention interval governs the shape of the recency gradient. Longer distractors push earlier observations further away in contextual space, steepening the gradient and increasing the dominance of the most recent observation; shorter distractors preserve contextual similarity, producing a flatter gradient and more balanced weighting. By varying distractor durations within subjects, we can test this prediction directly.

# 4 Method

## 4.1 Participants

We recruit participants aged 18–30 via Prolific (Peer et al., 2017). Inclusion criteria: native English speaker residing in the United States, no prior participation in related studies, and passing a comprehension check prior to the main task. Participants who fail the comprehension check after two attempts are excluded and replaced.

## 4.2 Design

The experiment crosses two between-subjects factors—process persistence ($\rho \in \{0.6, 0.9\}$) and encoding condition (passive observation vs. forced encoding)—with a continuous within-subject manipulation of distractor duration. Distractor duration is the primary experimental manipulation and is described in detail below. The encoding manipulation is included as a secondary factor: forced encoding ensures deep processing of each observation, whereas passive observation provides a closer analog to the Afrouzi et al. (2023) paradigm. The analyses reported in the Predicted Results section focus on the effects of distractor duration; we make no specific predictions regarding encoding condition.

### 4.3 Stimuli

Each participant sees a unique realization of a discrete-valued AR(1) process: $x_{t+1} = \text{round}[(1-\rho)\mu + \rho x_t + \varepsilon_t]$, with $\mu = 500$ and $\sigma_\varepsilon = 20$, constrained to $[1, 1000]$ and rounded to the nearest integer. The process begins at the unconditional mean, $x_1 = 500$. Using natural numbers simplifies forced encoding (participants type an integer), permits exact-match verification, and avoids rounding ambiguity. The scale accommodates the unconditional variability of the process—approximately $\pm 50$ around the mean at $\rho = 0.6$ and $\pm 100$ at $\rho = 0.9$—without risk of boundary effects.

We select $\rho = 0.6$ as a level where overreaction is robust in the Afrouzi et al. (2023) data, providing a strong baseline that our spacing manipulation can modulate. We select $\rho = 0.9$ because the rational forecast places substantial weight on $x_t$ (90%), creating a setting where the question is whether participants *over*-weight a legitimately informative observation.

### 4.4 Procedure

Each session consists of an observation phase and a forecasting phase (Figure 1). No cumulative graphical display of the time series is shown at any point; participants see only the current value on each trial and must retrieve all information about past observations from memory.

**Observation phase (20 trials).** Participants observe 20 realizations of the process, presented one at a time as a numerical value alongside a bar graphic indicating position on the 1–1000 scale (Figure 1A). In the forced-encoding condition, participants type the displayed value before proceeding; in the passive condition, they observe only. A lexical judgment distractor task of variable duration $d_t$ follows each observation, inducing context drift between successive encoding events. No forecasts are collected during this phase. The purpose is to give participants experience with the process—its scale, variability, and tendency to mean-revert—before forecasting begins.

**Forecasting phase (40 trials).** Each trial consists of two screens followed by a distractor (Figure 1B). On Screen 1, the participant enters a one-step-ahead forecast $F_t\, x_{t+1}$ as a natural number. On Screen 2, the actual value $x_{t+1}$ is revealed numerically alongside a bar graphic, with a red dot indicating the participant's forecast relative to the actual value. In the forced-encoding condition, participants type the actual value before proceeding; in the passive condition, they observe only. A distractor of variable duration $d_t$ then follows. Only the one-step-ahead forecast is collected on each trial, avoiding the multiple-elicitation confound identified in Section 2.

**Distractor task.** Between successive observations in both phases, participants perform a lexical judgment task for a variable duration $d_t$. On each distractor trial, a word drawn from the 1,638-word pool developed for the Penn Electrophysiology of Encoding and Retrieval Study (PEERS; Kahana et al., 2024) is presented, and participants judge either the word's size ("Will this item fit in a shoebox?") or its animacy ("Does this word refer to something living or not living?"), following the encoding tasks used in PEERS Experiments 1–3. A new word appears immediately after each response, and the task continues for the assigned duration. Participants are randomly assigned to either the size or animacy judgment for the entire session; this variation serves as a consistency check but is not the focus of the theoretical predictions. Because the distractor engages semantic processing that is distinct from the numerical forecasting task, it induces contextual drift without interfering with memory for the time series values.

**Distractor duration.** The duration of the distractor task is the primary experimental manipulation. On each trial, $d_t$ is drawn independently from a uniform distribution over $[d_{\min}, d_{\max}]$. We propose $d_{\min} = 2$ seconds and $d_{\max} = 12$ seconds. At 2 seconds, participants complete one to two lexical judgments—enough to engage the task and induce some context drift. At 12 seconds, participants complete approximately 6–8 judgments, producing substantially greater context drift. The range $[2, 12]$ was chosen to maximize the range of the log ratio $\log(d_{t-1}/d_t)$, the critical quantity under the ratio rule for recency (see Section 5.2). The mean distractor duration of 7 seconds yields a total session duration of approximately 20 minutes (including instructions), which is well suited to online data collection via Prolific.

Under retrieved-context theory, the amount of contextual drift between successive observations determines the steepness of the recency gradient. Longer distractors push earlier observations further away in contextual space, increasing the dominance of the most recent observation in the forecast; shorter distractors preserve contextual similarity across adjacent observations, producing a flatter gradient and more balanced weighting. By drawing distractor durations independently on each trial, we create trial-to-trial variation in the *relative* spacing of adjacent observations, providing a within-subject test of this prediction.

**Encoding manipulation.** In the passive-observation condition, participants view the actual value and forecast feedback on Screen 2 but make no explicit response to the observed value. Because no cumulative graph is available, any information not encoded during the Screen 2 presentation is lost. In the forced-encoding condition, participants type the integer value of $x_{t+1}$ into a text entry field after viewing the feedback; the typed value must match exactly, and mismatches prompt re-entry. This ensures deep processing of each observation. Forced encoding introduces a brief additional delay (1–2 seconds) between observation and distractor onset, which should be recorded via keystroke timestamps to allow post-hoc control.

### 4.5 Incentive Structure

We adopt the scoring function of Afrouzi et al. (2023): $S = 100 \times \max(0, 1 - |\Delta|/\sigma)$, where $\Delta$ is the forecast error and $\sigma = 20$. The cumulative score is displayed after each forecast. Base payment is \$3.00, with score-to-dollar conversion yielding expected incentive payments of approximately \$4.00, for total compensation of approximately \$7.00 over approximately 20 minutes.

### 4.6 Exclusion Criteria

Post-hoc exclusion criteria: (a) entering the same forecast on more than 80% of trials, (b) distractor-task accuracy below 60%, and (c) in the forced-encoding condition, requiring more than three re-entry prompts across the session.

### 4.7 Pre-registration

The experimental design, analysis plan, and exclusion criteria will be pre-registered prior to data collection.

## 5 Predicted Results

### 5.1 Baseline Overreaction

We first ask whether the overreaction documented by Afrouzi et al. (2023) persists—and potentially strengthens—when participants cannot consult a visual display and must forecast from memory alone. Following Afrouzi et al., we estimate the implied persistence $\rho_1^s$ by regressing the forecast on the current observation. We expect $\rho_1^s$ to exceed the true persistence $\rho$ at both levels, with the overreaction gap larger at $\rho = 0.6$ than at $\rho = 0.9$. A replication of this basic pattern establishes that the bias survives the removal of the graphical display, confirming that the phenomenon is not an artifact of visual attention to a time-series graph.

## 5.2 Distractor Duration Modulates Recency Weighting

To state the central prediction of retrieved-context theory precisely, it is helpful to borrow a distinction from the memory literature on the *ratio rule* for recency effects. In continuous-distractor free recall experiments—where participants study a list of items, each separated by a filled delay, and then recall as many items as possible after a final delay—the magnitude of the recency effect depends not on the absolute duration of the delays, but on the *ratio* of two intervals (Bjork and Whitten, 1974; Glenberg et al., 1980, 1983). The first is the *inter-presentation interval* (IPI): the duration of the distractor activity separating successive study items. The second is the *retention interval* (RI): the duration of the distractor activity between the last study item and the memory test. Recency increases approximately linearly with log(IPI/RI), a regularity that holds over time scales ranging from seconds to days (Glenberg et al., 1983; Nairne et al., 1997). Howard and Kahana (2002) showed that this scaling behavior emerges naturally from TCM's context-drift mechanism, and Howard (2004) derived the quantitative form of the prediction.

Our experiment has the same temporal structure. Consider trial $t$ of the forecasting phase (Figure 1B). The participant observes the actual value $x_{t+1}$ on Screen 2 and encodes it. A distractor of duration $d_t$ follows. When the next trial begins and the participant must forecast $x_{t+2}$, the *retention interval*—the delay between encoding the most recent observation and the moment of retrieval—is $d_t$. The *inter-presentation interval*—the delay that separated the encoding of $x_t$ from the encoding of $x_{t+1}$—is $d_{t-1}$ (plus a small constant for within-trial activity on trial $t$). Because both $d_t$ and $d_{t-1}$ are drawn independently on each trial from $[2, 12]$ seconds, the log ratio $\log(d_{t-1}/d_t)$ varies continuously across trials within each participant, providing a within-subject test of the ratio rule.

Under retrieved-context theory, the current state of temporal context serves as the cue for retrieving past observations. Context evolves according to $\mathbf{x}_t = (1-\zeta)\,\mathbf{x}_{t-1} + \zeta\,\mathbf{x}_t^{in}$, where $\zeta \in (0, 1)$ governs the rate of contextual drift (Howard and Kahana, 2002; Wachter and Kahana, 2024). A longer retention interval $d_t$ means that context has drifted further from the state in which the most recent observation was encoded—but this drift affects *all* past observations, pushing them uniformly further into contextual distance. A longer inter-presentation interval $d_{t-1}$, by contrast, selectively pushes the *second-most-recent* observation further from the most recent one. Thus the ratio $d_{t-1}/d_t$ controls the *relative* contextual accessibility of the two most recent observations, and hence the steepness of the recency gradient that shapes the forecast. Following Wachter and Kahana (2024), we set $\zeta = 0.35$, a value estimated from empirical fits to free recall data in their applications of retrieved-context theory to financial decisions.

Figure 2 displays the predicted weight on the most recent observation, $\beta_1$, as a function of $\log(d_{t-1}/d_t)$. The horizontal axis is the log ratio of the inter-observation interval to the retention interval—the same ratio-rule axis used to organize a half-century of findings on recency in free recall (Glenberg et al., 1980, 1983). The top axis shows concrete examples of distractor-duration pairs in seconds. The vertical axis is the predicted fraction of the forecast that is determined by the most recent observation; equal weighting of the two most recent observations corresponds to $\beta_1 = 0.5$.

The prediction is monotonic and approximately linear in the log ratio: as $\log(d_{t-1}/d_t)$ increases—meaning that the inter-observation gap is large relative to the retention interval—the most recent observation dominates the forecast more strongly, producing a steeper recency gradient and greater overreaction. At the right edge of the figure ($d_{t-1} = 12$ s, $d_t = 2$ s), the model predicts that the most recent observation receives roughly 79% of the weight. At the left edge ($d_{t-1} = 2$ s, $d_t = 12$ s), the weight drops to approximately 55%—still above equal weighting, because recency is never fully eliminated, but substantially flatter. This crossover prediction—that the *same participant*, forecasting the *same process*, will exhibit more or less overreaction depending on the recent history of distractor durations—is specific to retrieved-context theory and distinguishes it from accounts in which overreaction arises from a fixed cognitive bias, a representativeness heuristic, or a model-free reinforcement process, none of which predict that the temporal spacing of recent observations should modulate the magnitude of overreaction.

# 6 Experiment 2: Source-Context Reinstatement

Experiment 1 tests whether the temporal spacing of observations modulates recency weighting in forecasts—the ratio-rule prediction that follows from TCM's context-drift mechanism. We now turn to a second, complementary mechanism within the CMR framework: *source-context reinstatement*.

In the free recall literature, Polyn et al. (2009) demonstrated that when study lists are organized into trains of items encoded under the same orienting task (e.g., several consecutive size judgments followed by several consecutive animacy judgments), participants tend to recall items from the same encoding task in clusters—a phenomenon termed *source clustering*. CMR accounts for this pattern through the associative matrices $M^{FC}$ and $M^{CF}$, which bind item features to the temporal context that was active during encoding. When the encoding task changes, the contextual features associated with that task are reinstated, retrieving—as a cluster—the items that were previously encoded in the same task context (Polyn et al., 2009).

We ask whether this same mechanism operates in the forecasting setting. If the distractor

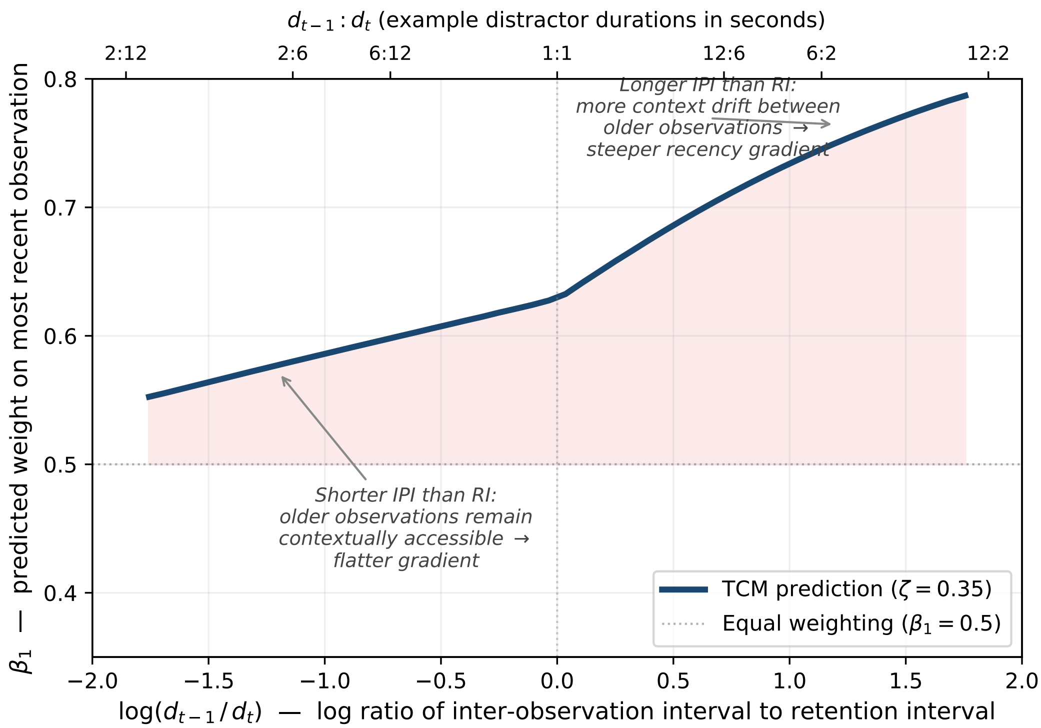


Figure 2: Predicted weight on the most recent observation ($\beta_1$) as a function of the log ratio of the inter-observation interval to the retention interval, under retrieved-context theory with $\zeta = 0.35$ (Wachter and Kahana, 2024). The horizontal axis follows the ratio-rule convention established by Glenberg et al. (1980): $\log(d_{t-1}/d_t)$, where $d_{t-1}$ is the distractor duration separating the two most recent observations (inter-presentation interval) and $d_t$ is the distractor duration between the last observation and the forecast (retention interval). The top axis shows example distractor-duration pairs in seconds. The shaded region indicates the range over which the most recent observation receives more than half of the total weight ($\beta_1 > 0.5$), corresponding to overreaction. Because both $d_t$ and $d_{t-1}$ are drawn independently from $[2, 12]$ seconds on each trial, the log ratio varies from $-1.79$ to $+1.79$ within each participant, providing a continuous within-subject test of the ratio-rule prediction.

task preceding each observation colors the encoding context of that observation, then a return to a previously experienced task type should reinstate the context associated with earlier encounters with that task. The reinstated context, in turn, should retrieve the AR(1) values that were encoded during those earlier encounters, biasing the forecast toward those values over and above what temporal recency alone would predict.

## 6.1 The Source-Reinstatement Prediction

Consider a participant who observes a sequence of AR(1) values, with the distractor task alternating in trains: several trials of size judgments, then several trials of animacy judgments, then a return to size judgments, and so on. During the first size train, each observation is encoded in a context that includes features of the size-judgment task. When the task switches to animacy, context drifts away from the size-task features and toward animacy-task features. When the task switches back to size, the size-task features are reinstated. Under CMR, this reinstated context serves as a retrieval cue that pulls back—via the $M^{CF}$ associations—the specific AR(1) values that were encoded during the prior size train.

The critical prediction is therefore a *contiguity effect in forecasting*: on the first trial after a task switch, the forecast should reflect a weighted average that includes values from the contextually matching prior train, not just the temporally recent values from the just-ended train. This prediction is specific to accounts that include source-context reinstatement. Alternative accounts of overreaction—fixed cognitive bias, representativeness, model-free reinforcement—predict no sensitivity to the identity of the distractor task, because they do not posit that the encoding context of past observations mediates their influence on the current forecast.

## 6.2 Method

The trial structure, platform, incentive structure, and exclusion criteria are identical to Experiment 1 unless noted below. One additional distractor is presented at the very start of the session, ensuring that every forecast—including the first—is preceded by a distractor whose task type colors the retrieval and encoding context.

**Design.** The experiment crosses one between-subjects factor—process persistence ($\rho \in \{0.6, 0.7\}$)—with a within-subject manipulation of distractor task type organized in trains. All participants perform the forced-encoding task; the passive-observation condition is not included.

**Process persistence.** We use $\rho \in \{0.6, 0.7\}$ rather than the $\{0.6, 0.9\}$ levels of Experiment 1. At $\rho = 0.9$, the autocorrelation of the AR(1) process at the lag separating a matching prior train from the current trial ($\rho^9 \approx 0.39$ for a typical separation of nine trials) is large enough that the mean of a prior same-task train predicts the current value for purely statistical reasons, confounding the reinstatement test. At $\rho = 0.7$ the corresponding autocorrelation is $0.7^9 \approx 0.04$, small enough that any predictive power of the matching-train mean can be attributed to memory retrieval rather than to the autocorrelation structure of the process. We retain $\rho = 0.6$ as the level where overreaction is most robust in the Afrouzi et al. (2023) data.

**Stimuli.** The AR(1) process uses the same parameterization as Experiment 1 ($\mu = 500$, $\sigma_\varepsilon = 20$, integer-valued on $[1, 1000]$), with identical parameters across task types. This ensures that any post-switch forecast bias must reflect context reinstatement, not a learned association between task type and process characteristics.

**Distractor task.** Both the size-judgment and animacy-judgment tasks are used within-subject, organized in alternating trains. Words are drawn from the same 1,638-word PEERS pool (Kahana et al., 2024). Within each train, the task type remains constant; at each switch boundary, the task changes. Train lengths are drawn independently from $\{3, 4, 5, 6\}$ trials, with the constraint that no two adjacent trains have the same length. This modest variability prevents participants from anticipating switches while keeping trains long enough to build compound source context—the same design logic used by Polyn et al. (2009), who employed trains of 2–6 items.

**Distractor duration.** In contrast to Experiment 1, where distractor duration varies on each trial ($d_t \in [2, 12]$ s), the distractor duration is held constant at $d = 4$ seconds on every trial. This isolates the source-context manipulation from the spacing manipulation tested in Experiment 1.

**Observation and forecasting phases.** The observation phase consists of 20 trials and the forecasting phase of 60 trials, both organized in alternating trains. The increase from 40 to 60 forecasting trials (relative to Experiment 1) provides approximately 12–18 task switches per session, yielding 8–14 critical "return" switches in which the participant encounters a task type for the second or subsequent time.

**Participants.** We recruit 150 participants per persistence level (300 total) via Prolific, using the same inclusion and exclusion criteria as Experiment 1.

**Pre-registration.** As in Experiment 1, the experimental design, analysis plan, and exclusion criteria will be pre-registered prior to data collection.

## 6.3 Predicted Results

**Baseline overreaction.** As in Experiment 1, we first confirm that participants overweight recent observations: the implied persistence $\rho_1^s$, estimated by regressing the forecast on the current observation, should exceed the true persistence $\rho$ at both levels.

**Post-switch forecast bias.** The central prediction of Experiment 2 concerns the first trials after each task switch. On these trials, the reinstated task context should retrieve values from the most recent same-task prior train, pulling the forecast toward those values over and above what temporal recency predicts.

To test this, we restrict attention to post-switch trials—those at within-train positions 0 and 1—that follow the participant's second or subsequent encounter with a given task type (so that a matching prior train exists). For each participant, we estimate the regression

$$F_t\, x_{t+1} = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \gamma\, \bar{x}_{\text{match}} + u_t, \tag{1}$$

where $\bar{x}_{\text{match}}$ is the mean of the AR(1) values observed during the most recent same-task prior train. CMR predicts $\gamma > 0$: the matching-train mean exerts a positive pull on the forecast even after controlling for the two most recent observations. We estimate $\gamma$ separately for each participant and test $\gamma > 0$ via a one-sample $t$-test across participants, separately at each persistence level. As a robustness check, we also report a specification that includes three recency lags ($x_t$, $x_{t-1}$, $x_{t-2}$).

Because the AR(1) parameters are identical across task types, any predictive power of $\bar{x}_{\text{match}}$ under the null hypothesis (no reinstatement) must arise from the residual autocorrelation of the process at the lag separating the matching train from the current trial. Simulations confirm that this residual autocorrelation is negligible at both $\rho = 0.6$ and $\rho = 0.7$ (false positive rates of approximately 5–7% at $N = 150$ per cell), ensuring that the test is well calibrated.

**Reinstatement decay across train position.** Under CMR, the reinstated context is strongest at the moment of the task switch and weakens as new within-train associations accumulate. We test this prediction using all trials within trains that have a matching prior train (not only the post-switch trials), estimating

$$F_t\, x_{t+1} = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \gamma_0\, \bar{x}_{\text{match}} + \gamma_1\, (p_t \times \bar{x}_{\text{match}}) + u_t, \tag{2}$$

where $p_t$ is the within-train position of trial $t$ (0 at the switch, 1 on the next trial, and so on). CMR predicts $\gamma_0 > 0$ (a positive reinstatement effect at the switch point) and $\gamma_1 < 0$ (decay of the effect as the participant accumulates new context within the current train). This analysis uses every trial in every train—not only the first one or two—and thus provides a more powerful test of the reinstatement mechanism.

**Persistence interaction.** At $\rho = 0.7$, within-train values are more autocorrelated than at $\rho = 0.6$: the process wanders less within a train, producing a more coherent signal for reinstatement to act on. We therefore predict a larger $\gamma$ at $\rho = 0.7$ than at $\rho = 0.6$, and test this interaction via a two-sample $t$-test comparing the participant-level $\gamma$ estimates across persistence conditions.

**Power.** Simulations indicate that with $N = 150$ per cell and approximately 27 post-switch observations per session, the post-switch analysis (Equation 1) achieves 80% power or greater to detect reinstatement effects as small as $\lambda = 0.08$—that is, conditions in which 8% of the forecast is attributable to the matching-train mean. The decay analysis (Equation 2), which uses all trials rather than only post-switch trials, is expected to be more powerful still.

# References


Afrouzi, H., Kwon, S. Y., Landier, A., Ma, Y., and Thesmar, D. (2023). Overreaction in expectations: Evidence and theory. *The Quarterly Journal of Economics*, 138(3):1713–1764.

Anderson, N. H. (1960). Effect of first-order conditional probability in a two-choice learning situation. *Journal of Experimental Psychology*, 59(2):73–93.

Anderson, N. H. (1964). Test of a model for number-averaging behavior. *Psychonomic Science*, 1(1–12):191–192.

Barberis, N. (2018). Psychology-based models of asset prices and trading volume. In Bernheim, B. D., DellaVigna, S., and Laibson, D., editors, *Handbook of Behavioral Economics*, volume 1, pages 79–175. North-Holland.

Barberis, N., Greenwood, R., Jin, L., and Shleifer, A. (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics*, 115(1):1–24.

Barberis, N., Greenwood, R., Jin, L., and Shleifer, A. (2018). Extrapolation and bubbles. *Journal of Financial Economics*, 129(2):203–227.

Barberis, N. and Jin, L. J. (2026). Model-free and model-based learning as joint drivers of investor behavior. Working paper, Yale University.

Barberis, N., Shleifer, A., and Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, 49(3):307–343.

Bjork, R. A. and Whitten, W. B. (1974). Recency-sensitive retrieval processes in long-term free recall. *Cognitive Psychology*, 6(2):173–189.

Bordalo, P., Gennaioli, N., La Porta, R., and Shleifer, A. (2019). Diagnostic expectations and stock returns. *The Journal of Finance*, 74(6):2839–2874.

Bordalo, P., Gennaioli, N., Ma, Y., and Shleifer, A. (2020). Overreaction in macroeconomic expectations. *American Economic Review*, 110(9):2748–2782.

Bordalo, P., Gennaioli, N., and Shleifer, A. (2018). Diagnostic expectations and credit cycles. *The Journal of Finance*, 73(1):199–227.

Bush, R. R. and Mosteller, F. (1955). *Stochastic models for learning.* John Wiley & Sons, Inc.

Coibion, O. and Gorodnichenko, Y. (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review*, 105(8):2644–2678.

Estes, W. K. (1950). Toward a statistical theory of learning. *Psychological Review*, 57:94–107.

Estes, W. K. (1957). Theory of learning with constant, variable, or contingent probabilities of reinforcement. *Psychometrika*, 22(2):113–132.

Estes, W. K. (1964). Probability learning. In Melton, A. W., editor, *Categories of Human Learning*, pages 89–128. Academic Press.

Fuster, A., Laibson, D., and Mendel, B. (2010). Natural expectations and macroeconomic fluctuations. *Journal of Economic Perspectives*, 24(4):67–84.

Gennaioli, N., Ma, Y., and Shleifer, A. (2016). Expectations and investment. *NBER Macroeconomics Annual*, 30:379–431.

Glenberg, A. M., Bradley, M. M., Kraus, T. A., and Renzaglia, G. J. (1983). Studies of the long-term recency effect: Support for a contextually guided retrieval theory. *Journal of Experimental Psychology: Learning, Memory & Cognition*, 12:413–418.

Glenberg, A. M., Bradley, M. M., Stevenson, J. A., Kraus, T. A., Tkachuk, M. J., and Gretz, A. L. (1980). A two-process account of long-term serial position effects. *Journal of Experimental Psychology: Human Learning and Memory*, 6:355–369.

Grant, D. A., Hake, H. W., and Hornseth, J. P. (1951). Acquisition and extinction of a verbal conditioned response with differing percentages of reinforcement. *Journal of Experimental Psychology*, 42(1):1–5.

Greenwood, R. and Hanson, S. G. (2013). Issuer quality and corporate bond returns. *The Review of Financial Studies*, 26(6):1483–1525.

Greenwood, R. and Shleifer, A. (2014). Expectations of returns and expected returns. *The Review of Financial Studies*, 27(3):714–746.

Hogarth, R. M. and Einhorn, H. J. (1992). Order effects in belief updating: The belief-adjustment model. *Cognitive Psychology*, 24(1):1–55.

Howard, M. W. (2004). Scaling behavior in the temporal context model. *Journal of Mathematical Psychology*, 48:230–238.

Howard, M. W. and Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3):269–299.

Humphreys, L. G. (1939). Acquisition and extinction of verbal expectations in a situation analogous to conditioning. *Journal of Experimental Psychology*, 25(3):294–301.

Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1):103–109.

Kahana, M. J., Lohnas, L. J., Healey, M. K., Aka, A., Broitman, A. W., Crutchley, P., Crutchley, E., Alm, K. H., Katerman, B. S., Miller, N. E., Kuhn, J. R., Li, Y., Long, N. M., Miller, J., Paron, M. D., Pazdera, J. K., Pedisich, I., Rudoler, J. H., and Weidemann, C. T. (2024). The Penn Electrophysiology of Encoding and Retrieval Study. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 50(9):1421–1443.

Kahana, M. J., Paron, J. D., and Wachter, J. A. (2025). Associative learning and representativeness. Working paper, University of Pennsylvania and Stanford University.

Kahneman, D. and Tversky, A. (1972). Subjective probability: A judgment of representativeness. *Cognitive Psychology*, 3(3):430–454.

La Porta, R. (1996). Expectations and the cross-section of stock returns. *The Journal of Finance*, 51(5):1715–1742.

Lohnas, L. J., Polyn, S. M., and Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2):337–363.

Malmendier, U. and Nagel, S. (2011). Depression babies: Do macroeconomic experiences affect risk taking? *The Quarterly Journal of Economics*, 126(1):373–416.

Murdock, B. B. (1962). The serial position effect of free recall. *Journal of Experimental Psychology*, 64(5):482–488.

Nairne, J. S., Neath, I., Serra, M., and Byun, E. (1997). Positional distinctiveness and the ratio rule in free recall. *Journal of Memory and Language*, 37:155–166.

Osth, A. F. and Fox, J. (2019). Are associations formed across pairs? a test of learning by temporal contiguity in associative recognition. *Psychonomic Bulletin & Review*, 26(5):1650–1656.

Peer, E., Brandimarte, L., Samat, S., and Acquisti, A. (2017). Beyond the Turk: Alternative platforms for crowdsourcing behavioral research. *Journal of Experimental Social Psychology*, 70:153–163.

Polyn, S. M., Norman, K. A., and Kahana, M. J. (2009). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1):129–156.

Shiller, R. J. (1981). Do stock prices move too much to be justified by subsequent changes in dividends? *American Economic Review*, 71(3):421–436.

Tversky, A. and Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131.

Wachter, J. A. and Kahana, M. J. (2024). A retrieved-context theory of financial decisions. *The Quarterly Journal of Economics*, 139(2):1095–1147.

Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press.